

3.

CREDIT SCORING METHODOLOGIES



Introduction

The methods used for credit scoring are evolving from traditional statistical techniques to innovative methods such as artificial intelligence, including machine learning such as random forests, gradient boosting, and deep neural networks.

Traditional Credit Scoring Methods

The most prominent techniques used to develop credit scorecards are statistical discrimination and classification methods. These include linear regression models, discriminant analysis, logit and probit models, and expert judgment-based models.

Linear Regression

Regression analysis is particularly useful in credit scoring because the statistical approach is easy to explain and predict risk parameters, such as the probability of default. In linear regression, the label (dependent variable or target outcome) is projected on a set of features (covariates or independent variables). Parameters that minimize the sum of squared residuals are chosen.

Discriminant Analysis

Discriminant analysis is a variation of regression analysis used for classification. The label is based on categorical data. The simplest variation is a label with two categories (for example, “default” versus “nondefault”). The original dichotomous linear discriminant analysis was developed by Sir Ronald Fisher in 1936 (Fisher 1936).

In default prediction, linear discriminant analysis was the first statistical method applied to systematically explain which firms entered bankruptcy, based on accounting ratios and other financial variables. Edward Altman’s 1968 model is still a leading model in practical applications (Altman 1968).

The original Altman Z-score model, developed using data of publicly held manufacturers, was as follows:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

where

X_1 = working capital / total assets

X_2 = retained earnings / total assets

X_3 = earnings before interest and taxes / total assets

X_4 = market value of equity / book value of total liabilities

X_5 = sales / total assets

Probit Analysis and Logistic Regression

For the dichotomous label in credit scoring, there have been several efforts to adapt linear regression methods to domains where the output is a probability value instead of any infinite real number. Many of such efforts focused on mapping the binary range to an infinite scale and then applying the linear regression on these transformed values.

In the probit model, an abbreviation for “probability unit,” the inverse standard normal distribution of the probability is modeled as a linear combination of the features. The logit function uses the log of odds, which is an abbreviation for “logistic unit” following the analogy for probit. In the logit model, the log of the odds ratio of the label is modeled as a linear combination of the features.

The logit model is a popular model for estimating the probability of default, because it is easy to develop, validate, calibrate, and interpret. Rather than choosing parameters that minimize the sum of squared errors (as in ordinary regression), estimation in logistic regression chooses parameters that maximize the likelihood of observing the sample values (see appendix A for technical specification).

Judgment-Based Models

Multiple methods may be employed to derive expert judgment-based models. One such a method is called the analytic hierarchy process (AHP), which is a structured process for organizing and analyzing complex decisions. The AHP model is based on the principle that when a decision is required on a given matter, consideration is given to information and factors, which can be represented as an information hierarchy. The decision makers decompose their decision problem into a hierarchical structure of more easily comprehended subproblems, each of which can then be independently analyzed. The key element of the AHP is that human judgments, not only the underlying information, be used to perform the evaluations. Human judgment is particularly critical in evaluating exceptions and instances that do not have precedence or are significantly underrepresented in the data.

For example, Bana e Costa, Barroso, and Soares (2002) developed a qualitative credit scoring model for business loans based on concepts of the AHP.

Artificial Intelligence and Machine Learning in Credit Scoring

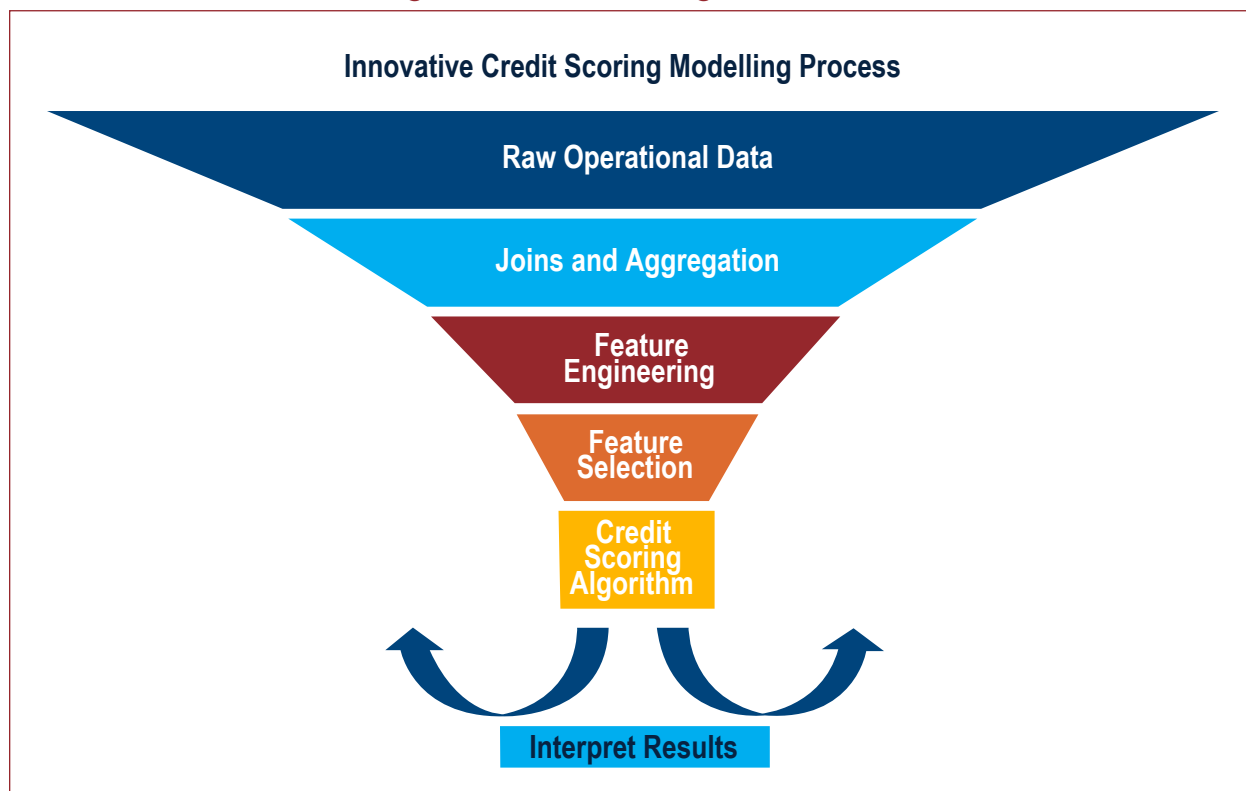
Artificial intelligence (AI) is the application of computational tools to address tasks traditionally requiring human sophistication (SAS 2019). AI enables machines to learn from experience, adjust to new inputs, and perform human-like tasks (FSB 2017). Most AI examples that are popular today—from self-driving machines to superhuman doctors—rely heavily on deep learning and natural language processing. These techniques leverage the ability of computers to perform tasks, such as computer vision and chatbots, by learning from experience.

Today’s evolving AI is made possible by rapid development in foundational technologies such as computing power, data, and innovative algorithms. Using these technologies, computers can be trained to accomplish specific tasks by processing and recognizing patterns in the data, while the data may be of different types and from different sources.

AI is a broad field, and machine learning is a subcategory. Machine learning is defined as a method of designing a sequence of actions to solve a problem, known as an algorithm, which optimize automatically through experience with limited human intervention (SAS 2019). These techniques can be used to find complex patterns in large amounts of data from increasingly diverse and innovative sources (SAS 2019).

Deep learning is a form of machine learning that uses algorithms that work in layers inspired by the structure and function of the human brain (SAS 2019). Deep learning algorithms can be used for supervised, unsupervised, or reinforcement learning. Recently, deep learning has led to remarkable results in fields such as image recognition and natural language processing. For example, deep learning may be used to classify images, recognize speech, detect objects, and describe content. Voice recognition systems are powered, in part, by deep learning.

Box 3.1: Machine Learning for Credit Scoring



At a high level, the application of machine learning algorithms for credit scoring involves the following high-level process that can be split further into multiple subphases:

- Access raw data.
- Combine, join, and aggregate input data.
- Engineer features, either manually using expert input or using automated approaches.
- Select useful features.
- Apply the machine learning algorithm to the training data set.
- Interpret and assess results (see figure 3.1).
- Some techniques include a feedback loop where the algorithm learns from experience.

Innovative credit scoring methods include supervised, unsupervised and semi-supervised learning techniques.

Supervised Learning Techniques

In supervised learning, the algorithm is developed using data that contain a label (dependent variable or event) and independent features (variables). The algorithm then predicts future or unknown values of the label of interest, using features (independent variables). For instance, a data set of counterparties may contain labels on some data points identifying those that are in default and those that are not in default. The algorithm will learn a general rule of classification that it will use to predict the labels for other observations in the data set.

Some of the supervised techniques include regression, decision trees, random forests, gradient boosting and deep neural networks.

Decision trees

Decision trees are typically schematic, tree-shaped diagrams used to show a statistical probability. Classification and regression trees (CART) are one of the most well-established supervised learning techniques. CART works by repeatedly finding the best feature to split the data into subsets. The partition improves the isolation of the label with each split.

Decision trees can be used for either classification, for example, to determine the category of an observation (that is, default or nondefault), or for prediction, for example, to estimate a numeric value (that is, the loss given default).

Random forests

Random forests are a combination of tree predictors such that each tree depends on a sample (or subset) of the model development data (or training data) selected at random (Breiman 2001). Working with multiple different subdata sets can help reduce the risk of overfitting. Random forests or random decision forests are ensemble methods for regression and classification problems based on constructing a multitude of decision trees and outputting the class that may be either the mode of the classes (classification) or the mean prediction (regression) of the individual trees (see appendix A for technical specification).

Gradient boosting

Gradient boosting is an ensemble method for regression and classification problems. Gradient boosting uses regression trees for prediction purposes and builds the model iteratively by fitting a model on the residuals. It generalizes by allowing optimization of an objective function.

Deep neural networks

Instead of organizing data to run through predefined equations, deep neural networks train the algorithm to learn on its own by recognizing patterns using many layers of processing.

Deep neural networks, also referred to as deep-learning networks, are neural networks with additional hidden layers. Each layer of nodes trains on a set of features based on the previous layer's

output. Hence, progression through each layer results in increasingly complex layers of features that are aggregated, recombined information learned from the previous layer.

This characteristic makes deep-learning networks able to identify highly complex nonlinear patterns with large volumes of data and dimensions (Press 2017). On the flip side, deep neural networks are also prone to overfitting (see appendix A for technical specification). Overfitting may be reduced by validating a new model on an out-of-sample data set.

Other popular techniques include support vector machines (see appendix A for technical specification).

Unsupervised Learning Techniques

Unsupervised learning refers to methods where the data provided to the algorithm do not contain labels (events). The algorithm is required to detect patterns in the data by identifying clusters of observations that demonstrate similar underlying characteristics, for example. In other words, these algorithms serve to explore the properties of the data examined, rather than to predict new or unknown data. Unsupervised techniques include clustering, K-means clustering, and hierarchical clustering.

Clustering

Clustering is the process of obtaining natural groups from the data. These are groups and not classes, because, unlike classification, instead of analyzing data labeled with a class, clustering analyzes the data to generate this label. The data are grouped on the basis of the principle of maximizing the similarity between the elements of a group by minimizing the similarity between different groups. That is, groups are formed such that the objects of the same group are very similar to each other and, at the same time, they are very different from the objects of another group.

Clustering algorithms are descriptive rather than predictive. For example, a clustering algorithm may be used to look for a borrower that has characteristics similar to a borrower that is difficult to assess. If

the algorithm finds an appropriate cluster for the borrower, the average default assessment of the cluster may be used as an estimate of the default assessment of the borrower.

This report distinguishes between two types of clustering methods: K-means and hierarchical clustering.

K-means clustering

K-means clustering aims to partition a set of observations into K clusters, resulting in the partitioning of the data into groups. It starts by placing K central positions (centroids) in random locations in the multidimensional (Euclidian) space. It then uses the Euclidean distance between data points and centroids to assign each data point to the cluster that is closest. The process is iterative (see appendix A for the technical specification).

Hierarchical clustering

Hierarchical clustering starts by assigning all data points as their own cluster. Based on the Euclidian distance, it then combines the two nearest data points and merges them together to form a new cluster. The process continues until convergence.

Other Techniques Related to Credit Scoring Methods

Automated Feature Engineering

The success of machine learning models highly depends on the success of feature engineering. Feature engineering is the process of transforming the input data set into features to better understand the underlying structures in the data and improve model accuracy.

The process may be highly iterative: multiple sets of features may be created and evaluated in several phases. Thousands or even millions of feature candidates may be generated; thus, there is a need to intelligently select relevant candidates and to avoid overfitting in subsequent model development.

Owing to its manual nature, the traditional approach to feature engineering tends to be tedious, time consuming, and error prone. The emergence of automated feature engineering seeks to address these shortcomings and allow analysts to spend their time more efficiently on other aspects of the modelling process (Koehrsens 2018).

One such example is the deep feature synthesis algorithm, which automatically performs feature engineering on relational and multitable data (Kanter and Veeramachaneni 2015). The algorithm captures features by stacking multiple primitives. Feature primitives are basic mathematical operators that can be applied to raw variables as aggregations and transformations. An example of a transformation is the ratio of two raw variables such as debt and income. Primitives are used to form new features by applying them across data sets or stacking them to create more complex features.

Reinforcement Learning

Reinforcement learning is an emerging method that falls between supervised and unsupervised learning. It has been applied in credit processes such as collections.

Reinforcement learning allows the machine to learn behavior based on feedback from the environment. This behavior can be learned once or adapted as time goes by (Champanand n.d.).

The goal is to train an algorithm that considers the environment, takes actions, and receives feedback from the environment in terms of rewards (the feedback mechanism) to learn which optimal actions to take. The ideal behavior maximizes the reward. Reinforcement learning is popular in robotics and game theory.

This automated learning scheme implies that there is little need for human intervention.

Natural Language Processing

Natural language processing (NLP) is the ability of machines to analyze, understand, and generate human language, including speech. It is a branch of artificial intelligence that helps computers understand, interpret, and manipulate human language. NLP draws from many disciplines, including computer science and computational linguistics. For text, NLP often uses semantics, concept identification, and sentiment analysis. The next stage of NLP is natural language interaction, which allows humans to communicate with machines using normal, everyday language to perform tasks (SAS 2019). NLP capabilities can be extended to natural language understanding, which is how a machine tries to understand the meaning of spoken or written language. This field goes beyond just the structural understanding of language into the ability of a system to interpret intent, resolve context and word ambiguity, and even generate well-formed language on its own.

Blockchain for Decentralized Credit Scoring

A blockchain is a distributed ledger in which any computer that is part of a network can participate. For data to be updated on the blockchain, they require verification from multiple sources.

A main benefit of the blockchain network is that it lacks centralized points of vulnerability for hackers to exploit, reducing cybersecurity risks. Blockchain

security methods use encryption technologies. This feature enables blockchain technology to provide a much higher level of security in storing information (“What Is a Blockchain?” 2018). Several financial technology providers have leveraged this process to provide users a more secured means to retrieve their credit score, as well as share credit history with CSPs.

Decentralized credit scoring takes thousands of features of a user’s credibility from multiple data sources, while credit scores are calculated via decentralized credit scoring algorithms. The features may consider data from social media to online shopping history to create a universal digital identity and a comprehensive credibility profile owned by the individual (“Colendi Technical Paper” n.d.; box 3.1). However, blockchain is not yet widely adopted for credit scoring. A significant issue with blockchain is that there is yet no significant scalability and that the costs per transaction tend to rise steeply.

Understanding and Interpreting Credit Scoring Models

Introduction

One major concern raised by regulatory and industry bodies (Monetary Authority of Singapore n.d.; Foy 2018) is the ability for humans to interpret, understand, explain, and justify the decisions made by credit scoring methods using large numbers of variables, particularly using new, innovative approaches.

Box 3.1: Use Case—Colendi

The Colendi platform is as follows:

- Colendi is a decentralized microcredit platform that uses machine-learning algorithms to generate a credit score and identity check based on alternative data sources while ensuring safety and privacy. Colendi can help provide services to the underbanked and unbanked.
- Users authorize the Colendi protocol to read data related to their smartphones, social media, purchases, and more than 1,000 personal data points to be used in credit scoring machine-learning models.
- The credit scoring machine-learning models are designed to analyze the data through integrated blockchain nodes without changing or retaining the data, ensuring that any data that are available to the application are not moved to any servers or revealed to any parties.

Source: “Colendi Technical Paper” n.d.

The European Banking Authority (EBA) and other regulatory bodies highlighted the need for further research to improve the interpretability of the models to mitigate risks such as bias and discrimination (European Commission 2018a).

Interpretability can also be defined as tackling the challenge of asserting a causal connection between input data to models and the way that input data impact model output and decisions (Tan 2018).

Interpretability of Model Inputs

Understanding of the input data is important to strengthen trust and to ensure that personal data are lawfully used and the risk of amplifying bias in historical data is addressed. Interpretability of the model inputs can be increased by documenting input data, with sufficient tracking of permission where applicable; utilizing exploratory data analysis; and ensuring the selection of meaningful features is justified and proper governance of feature selection is in place. Highly engineered features are useful to improve model accuracy; however, they may not be interpretable when used in the models.

Interpretability of Modelling Logic

Traditional approaches like regression and decision tree algorithms are interpretable and explainable when used with few features. The models typically use interpretable transformations and show intuitive relationships, which help users understand the models.

Linear models are particularly easy to interpret because the relationship between the features and the label is modelled in a linear function. Linear models can also be used to effectively capture nonlinear relationships (for example, exponential relationships) by transforming raw variables. The linearity of the

learned relationship makes the interpretation easy and straightforward.

Decision trees are also easy to interpret; by starting from the root node, one can trace the classification logic through the rule sets. The leaf nodes inform the label.

Deep neural networks, random forests, and gradient boosting machines are regularly considered as opaque or black box models (FSB 2017). These algorithms use many features and complex transformations, making it challenging to interpret the relationship between the feature and the target.

Interpretability of Postmodelling Results

Some research studies (FSB 2017) consider the interpretation of the model predictions to help inspect the dynamics between input features and output predictions. Postmodelling results analysis may be necessary in cases where, for example, the machine learning model is opaque.

Postmodelling interpretability tests can help in the understanding of the most important features of a model: how those features affect the predictions, how each feature contributes to the prediction, and how sensitive the model is to certain features.

Model-agnostic techniques include partial dependence (PD) plots, individual conditional expectation (ICE) plots, and local interpretable model-agnostic explanations (LIME) (Champanand n.d.). See appendix B for specification.

Model-specific techniques include variable importance output from random forest, for example.